

HSBC Document Intelligence Platform

Next-Generation Document Processing for Insurance Underwriting

Technical Deep Dive

Agenda

1. **Current State vs. Platform Capabilities**
2. **Self-Service Document Type Onboarding**
3. **AI-Powered Evaluation Framework**
4. **Pluggable Architecture**
5. **Human-in-the-Loop Workflows**
6. **Extensible Orchestration**
7. **Live Demo**
8. **Future Extensibility**

Current Pilot vs. Platform

Capability	Current Pilot	This Platform
Document Extraction	✅ OCR + GPT	✅ OCR + GPT (Multiple Models)
Output Format	JSON Key-Value	JSON + Citations + Confidence
Evaluation	⚠️ Limited	✅ AI-Powered (Correctness + Completeness)
New Document Types	❌ Developer Required	✅ Self-Service Onboarding
Quality Assurance	❌ Manual Review	✅ Automated Scoring
Architecture	Monolithic	Pluggable Components
Workflow Extensibility	Limited	Full Orchestration Engine

The Challenge: Adding New Document Types

Current Process

```
graph TD; A[New Document Type Request] --> B[Developer writes custom extraction logic]; B --> C[Manual testing with sample documents]; C --> D[No way to measure accuracy]; D --> E[Deploy and hope it works]; E --> F[Issues discovered in production];
```

Timeline: Weeks to months per document type

The Solution: Self-Service Onboarding

4-Step Wizard for Business Users

Step 1: Upload & Classify

- Upload sample document
- Auto-classification with confidence
- Optional ground truth

Step 2: Configure

- Select extraction model
- Define field schema
- Customize prompts

Step 3: Test & Evaluate

- Real-time extraction preview
- AI-powered quality scores
- Iterate until satisfied

Step 4: Finalize

- One-click deployment
- Version control
- Instant availability

AI-Powered Evaluation Framework

The Problem

How do you know extraction is correct?

- Without ground truth data, accuracy is unknown
- Issues discovered only in production
- Manual review of every extraction is not scalable

Our Solution: Dual AI Evaluators

Evaluator	Purpose
Correctness	Is the value present in source document?
Completeness	Is the full value captured?

Evaluation: No Ground Truth Required

How It Works

Correctness Evaluator

- Tokenizes extracted value and source text
- Measures token-level coverage ratio
- 100% = all tokens found in source

Completeness Evaluator

- Uses LLM to assess extraction quality
- Checks if full context was captured
- Provides reasoning for scores

Evaluation in Action

Per-Field Quality Scores

Field	Extracted Value	Correctness	Completeness	Status
Applicant Name	John Smith	95% ●	100% ●	✓ Pass
Policy Amount	\$500,000	100% ●	100% ●	✓ Pass
Coverage Start	2026-01-15	88% ●	92% ●	✓ Pass
Medical History	Diabetes Type 2	72% ●	65% ●	⚠ Review

Scores are color-coded: ● $\geq 80\%$ | ● 50-79% | ● $< 50\%$

Each field links to source text for visual verification

Pluggable Architecture

Swap Components Without Code Changes

Layer	Current	Can Add
Classification	Azure CU	Custom ML
Extraction	GPT-4o, GPT-4V	Claude, Gemini
Summarization	GPT-4o	Any LLM
OCR	Azure Doc Intel	Tesseract

How Pluggable Architecture Works

Interface-Driven Design

- Each component implements a **Python Protocol**
- Swap implementations via **configuration**
- Register new models in **Model Registry**
- **No redeployment** needed for new models

Benefits

- Future-proof: adopt new AI models easily
- Vendor flexibility: avoid lock-in
- Test alternatives: compare performance

Human-in-the-Loop

Smart Routing Based on Confidence

Confidence	Action	Effort
● High ($\geq 80\%$)	Auto-approve	None
● Medium (50-79%)	Quick review	Minimal
● Low ($< 50\%$)	Full review	Full

Result: Reviewers focus only on uncertain extractions

Citation Highlighting

Visual Verification in Seconds

- **Click any field** → Source highlighted in PDF
- **Side-by-side view:** Data + Document
- **Approve or Edit** directly in UI
- **Audit trail:** All changes logged

Extensible Orchestration

Define Custom Workflows

```
workflow = (  
    PipelineBuilder()  
    .add_step("ingest", IngestDocument())  
    .add_step("classify", ClassifyDocument())  
    .add_step("extract", ExtractEntities())  
    .add_step("validate", BusinessRules())  
    .add_step("summarize", GenerateSummary())  
    .build()  
)
```

Orchestration Capabilities

Built-in Features

Capability	Description
Retry Logic	Automatic retry with backoff
Error Handling	Graceful fallbacks
Audit Logging	Full traceability
Async Execution	Non-blocking processing
Component Discovery	Auto-register new steps

Architecture Benefits

For Development

- **Monorepo Structure:** Single source of truth
- **Typed Interfaces:** Python Protocols
- **Independent Testing:** Per-component tests
- **Version Control:** Schema versioning

For Operations

- **Azure Native:** Cosmos DB, Blob Storage, OpenAI
- **Observability:** Trace IDs, metrics, logs
- **Scalability:** Stateless components
- **Security:** Managed identities, RBAC

Demo Walkthrough

Demo 1: Self-Service Onboarding

Upload → Configure → Test → Evaluate → Deploy

Demo 2: Extraction Results

Field scores → Citation links → PDF overlay

Demo 3: Human Review

Flagged items → Verify → Approve/Edit

Beyond Insurance Underwriting

The Platform is Use-Case Agnostic

Use Case	Document Types	Key Fields
Insurance Underwriting	Applications, Medical Reports	Applicant info, Coverage, Conditions
Claims Processing	Claim Forms, Invoices, Reports	Claim details, Amounts, Dates
Credit Insurance	Financial Statements, Credit Reports	Exposures, Ratings, Terms
Trade Finance	Letters of Credit, Bills of Lading	Parties, Amounts, Shipping details
KYC/AML	ID Documents, Utility Bills	Identity verification, Address proof

Same platform, different configurations

Roadmap: Designed for Scale

Ready for Future Requirements

Capability	Status
Multi-language (Chinese)	Architecture ready
Multi-document scenarios	Designed for
Real-time processing	Event-driven ready
Legacy integration	API-first design

Roadmap: Pluggable Extensions

Easy to Add

- **New AI Models:** As they emerge (GPT-5, etc.)
- **Business Rules:** Custom validation engines
- **Industry Validators:** Domain-specific checks
- **Data Enrichment:** External API integration
- **Fraud Detection:** Pattern recognition modules

Key Differentiators Summary

Feature	Business Value
Self-Service Onboarding	Days → Hours for new document types
AI Evaluation	Quality assurance without labeled data
Pluggable Architecture	Future-proof, swap components easily
Human-in-the-Loop	Smart routing reduces review burden
Extensible Orchestration	Adapt workflows to any business process
Citation Linking	Visual verification of extracted data

Questions?

Ready for Live Demo

Appendix: Technical Stack

Layer	Technology
Frontend	Vue 3, Vuetify, TypeScript
Backend API	FastAPI, Python 3.11+
AI/ML	Azure OpenAI (GPT-4o, GPT-4V), Azure Document Intelligence
Database	Azure Cosmos DB
Storage	Azure Blob Storage
Infrastructure	Azure Bicep (IaC)
Testing	Pytest, Playwright

Appendix: Evaluation Metrics Detail

Correctness Evaluator

- Tokenizes extracted value and source text
- Measures token-level coverage
- Score = matched tokens / total extracted tokens
- 100% = all extracted tokens found in source

Completeness Evaluator

- Uses GPT-4 to assess extraction quality
- Checks if full context was captured
- Identifies missing information
- Provides reasoning for scores